

Day 1:

Intro: (25 minutes)

- Introduce myself & Syed's Introduction.
- Talk about the class being blended and that our first activity today is to help demonstrate how these sessions will run. (currently looking at 4 groups of sizes, 3,3,2,2)

Task: The four 4's: Use four 4's and the operations of +, -, *, & / to make the #'s 1-20.

On Board:

- Acceptable Operations: +, -, x, ÷
- Example: $(4 \times 4) + 4 - 4 = 16$

$1 = \frac{4}{4} + 4 - 4$	$11^* = 9 + \frac{9+9}{9}$
$2 = \frac{4}{4} + \frac{4}{4}$	$12 = (4-1) \times 4$
$3 = \frac{4+4+4}{4}$	$13^* = 9 + \frac{9}{9} + \sqrt{9}$
$4 = (4-4)4 + 4$	$14 = (4+4+4) + \sqrt{4}$
$5 = \frac{(4 \times 4) + 4}{4}$	$15 = (4 \times 4) - \frac{4}{4}$
$6 = \frac{4+4}{4} + 4$	$16 = (4 \times 4) + 4 - 4$
$7 = (4+4) - \frac{4}{4}$	$17 = (4 \times 4) + \frac{4}{4}$
$8 = (4+4) + 4 - 4$	$18 = (4 \times 4) + 4 - \sqrt{4}$
$9 = (4+4) + \frac{4}{4}$	$19^* = 9 + 9 + \frac{9}{9}$
$10 = 4 + 4 + 4 - \sqrt{4}$	$20 = 4 \times \left(4 + \frac{4}{4}\right)$

During the activity:

- Require students to cycle through who uses the marker each time a new number is found.

Extensions:

- What can you make?
- What can't you make?
- Can you make any of the numbers you couldn't make originally if you added the square root operation?
- Can you make any of the remaining numbers you haven't been able to make if you used a number besides 4 along with the square root?

(5minutes) Discussion:

- This activity isn't content based, but it highlights some important things about our class and these sessions
 - We won't always know a solution at the onset of a problem.
 - Working with others will help us to see things that we might have missed.
 - Learning to express our thoughts to others can be difficult.

(10minutes) Syllabus:

- Cover major grading points, homeworks, exam, worksheets, and Blackboard. The rest can be left for them to read.
- SI sessions.

(35minutes) Content:

- Introduce the Graphs to Know.
- Show them the Skills Checks and state that we are focused for the remainder of the day on #1
 - They can cover remaining topics there on their own using the worksheets.
- Cover piecewise functions from the Skills check worksheets (with some modifications and extensions)
 - Q1: $f(x) = \begin{cases} x & \text{if } x \geq 2 \\ x^2 - 3 & \text{if } 0 \leq x < 2 \\ 5 & \text{if } x < 0 \end{cases}$
 - Extension: If the linear pieces of this function had their same starting points, but were quadratic, what could this new function's formula be?
 - Q2: $f(x) = \begin{cases} 3x - 5 & \text{if } x > 4 \\ \ln(x) & \text{if } 0 < x < 4 \end{cases}$
 - Extension: What would the function look like if we added the following?
$$f(x) = \begin{cases} 3x - 5 & \text{if } x > 4 \\ \ln(x) & \text{if } 0 < x < 4 \\ e^x & \text{if } x \leq 0 \end{cases}$$
 - Is there a problem with have a point on the vertical asymptote?

(10 minutes) Conclusion:

- Reiterate goals of the sessions
- Discuss Blackboard.